

DevOps Online Instructor-Led Training

Program Overview

This DevOps Program will provide you with in-depth knowledge of various DevOps tools including Git, Jenkins, Docker, Ansible and designed in a way to help you gain knowledge through best practices in CICD (Continuous Integration & Continuous Deployment) and finally, Continuous Monitoring of software throughout its development life cycle.

Pre-requisites

Learners should have a technical background with an understanding of Linux fundamentals

Target Audience

This DevOps Online Training will be suitable for software developers, technical project managers, architects, operations support, deployment engineers, Freshers and Induction Batch

Training Key Features

- Live, online classroom training by experienced instructor
- Real-World Projects: Gain practical, hands-on experience
- Training involves whiteboard and Practical Demos
- Flexible and adjustable class timings on weekday or Weekends
- Tailored modules to boost your Dev+Sec+Ops expertise

Course Duration

- Duration: 1 Month

Course curriculum

1. Waterfall Model (Traditional)

2. DevOps Essentials

- Why DevOps?
- What is DevOps?
- DevOps Market Trends
- What is a Pipeline?

3. Introduction to Agile

- Agile Methodology
- Daily Stand-ups

4. GitHub Branching Strategy

5. Developer Workflow on GitHub

6. Managing Source Control with Git

- What is version control
- Central and Distributed Version Control
- What is Git?
- Developer/coding workflows
- Branching and merging Concepts
- Working with Remote Repositories
- **Demo**

7. Understanding and working with Maven Build Tool - Eclipse IDE

- Overview of Various Build tools
- What is Maven
- Maven Plugins
- Maven Archetypes
- Project Object Model (POM)
- Source Control Integration
- Maven build lifecycle
- **Demo**

8. Setup Virtual Machines with VMware, Jenkins Master and Slave node

9. Continuous Integration (CI) Pipeline with Jenkins on Virtual Machine

- What is CI (Continuous Integration)?
- Why is CI Required?
- Introduction to Jenkins
- Jenkins Architecture
- Adding a slave node to Jenkins Master
- Declarative Pipeline using Pipeline as a Code (Jenkinsfile)
- Webhooks
- **Demo**

10. Containerization using Docker

- Introduction to Docker and Container
- Understanding of images and containers

- Introduction to Docker Networking
- Container Life Cycle
- **Demo**

11. Continuous Deployment on Docker using Jenkins

- **Demo**

12. [Optional] Introduction to Kubernetes

- Basics of Kubernetes container orchestration
- Differences between Docker Swarm and Kubernetes
- Kubernetes Architecture
- Installing Kubernetes using Kubeadm
- Creating Pods and Deployments using YAML
- Selectors & Labels in Kubernetes
- Using ReplicaSets & Rolling Updates
- Scheduling the applications on the container
- Services in Kubernetes
- **Demo**

13. Configuration Management with Ansible

- Overview of Configuration Management
- Introduction to Ansible
- Ansible Architecture
- Ansible Components
- Installation & Configuration
- Writing Ansible Playbooks
- Working with Ansible variables
- Working with Ansible Modules
- **Demo**

Caption Project

- Freestyle Jenkins job to build and compile Java project using Maven build tool
- Declarative pipeline in Jenkins to build Java application using docker image and then push the image to Dockerhub